

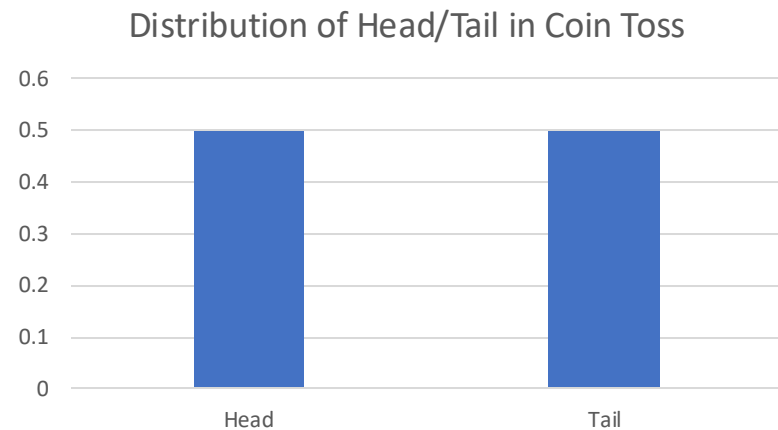


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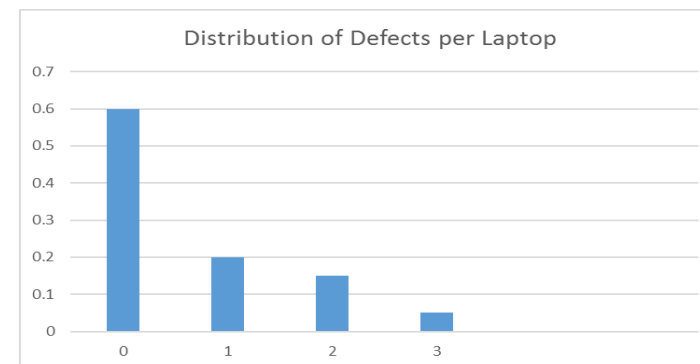
Business Analytics with GenAI

Session 2: Introduction to Data Distributions

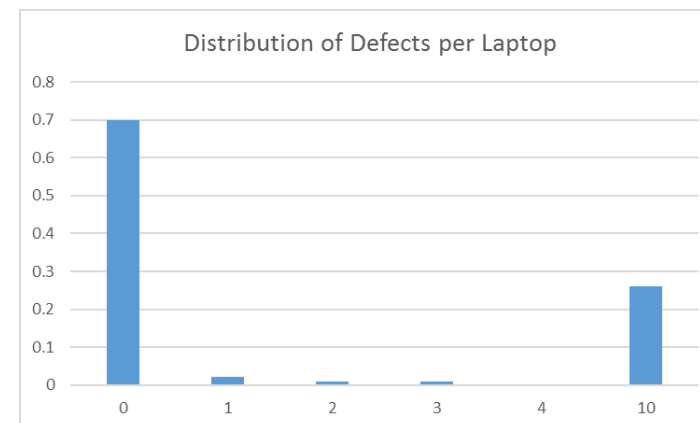
Data may be distributed in different ways



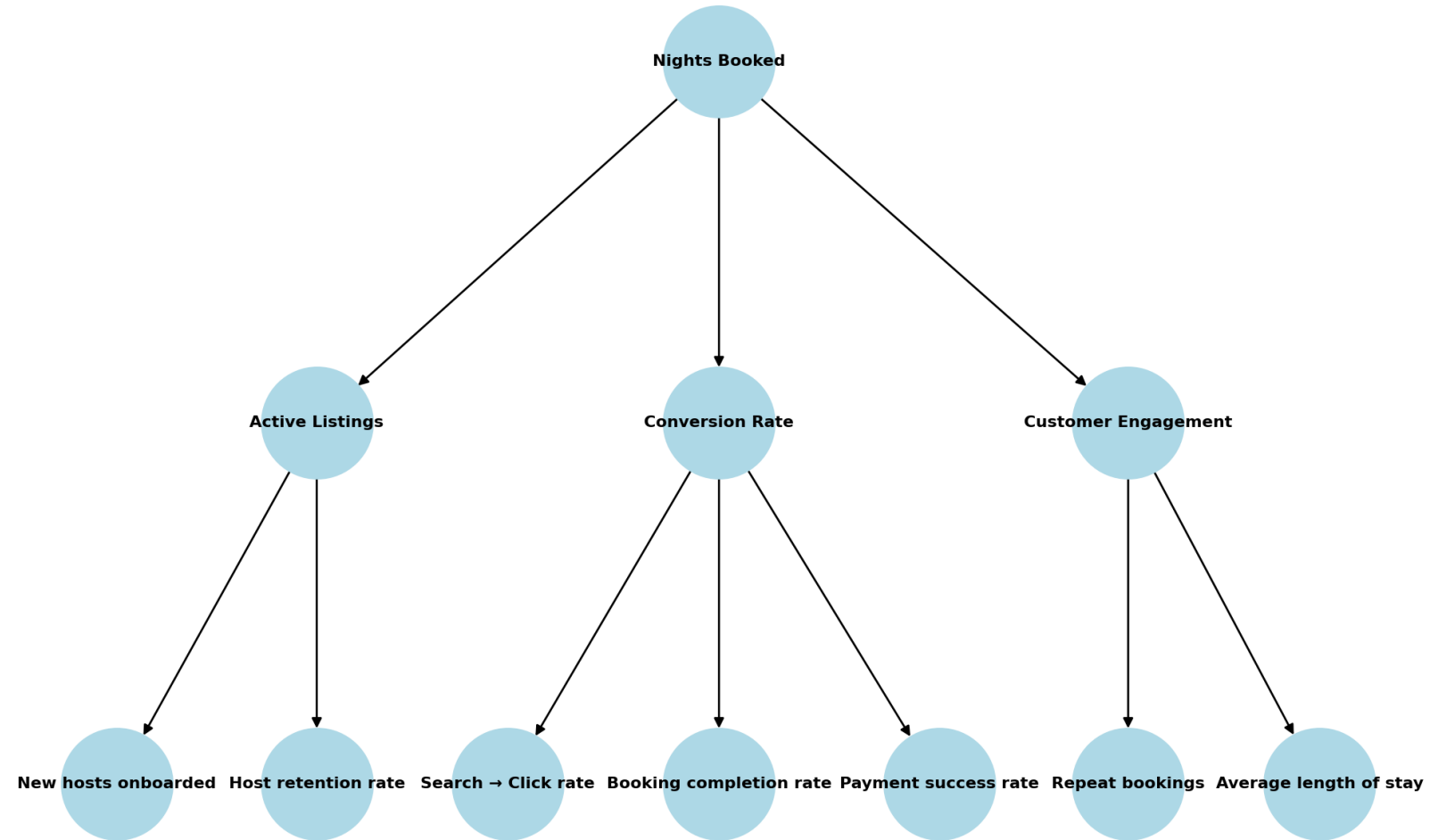
Factory 1:



Factory 2:



AirBnB KPI Tree

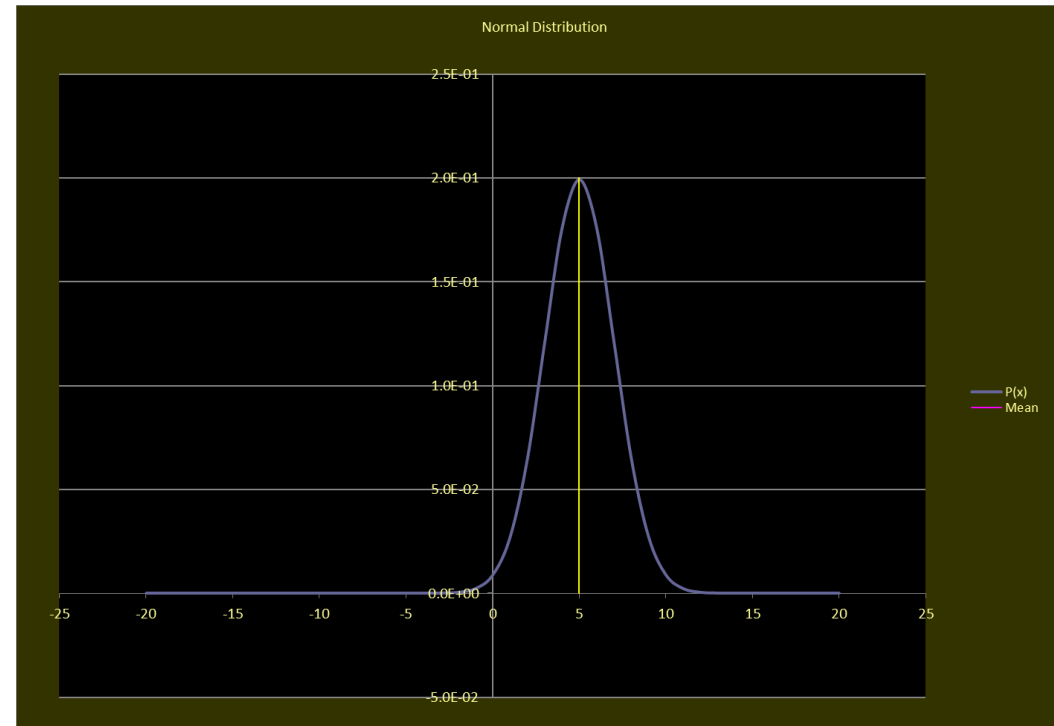


Normal Distributions

- Bell shaped
- Symmetric distribution (mean and median are the same)
- Two parameter distribution
 - Mean and Standard Deviation

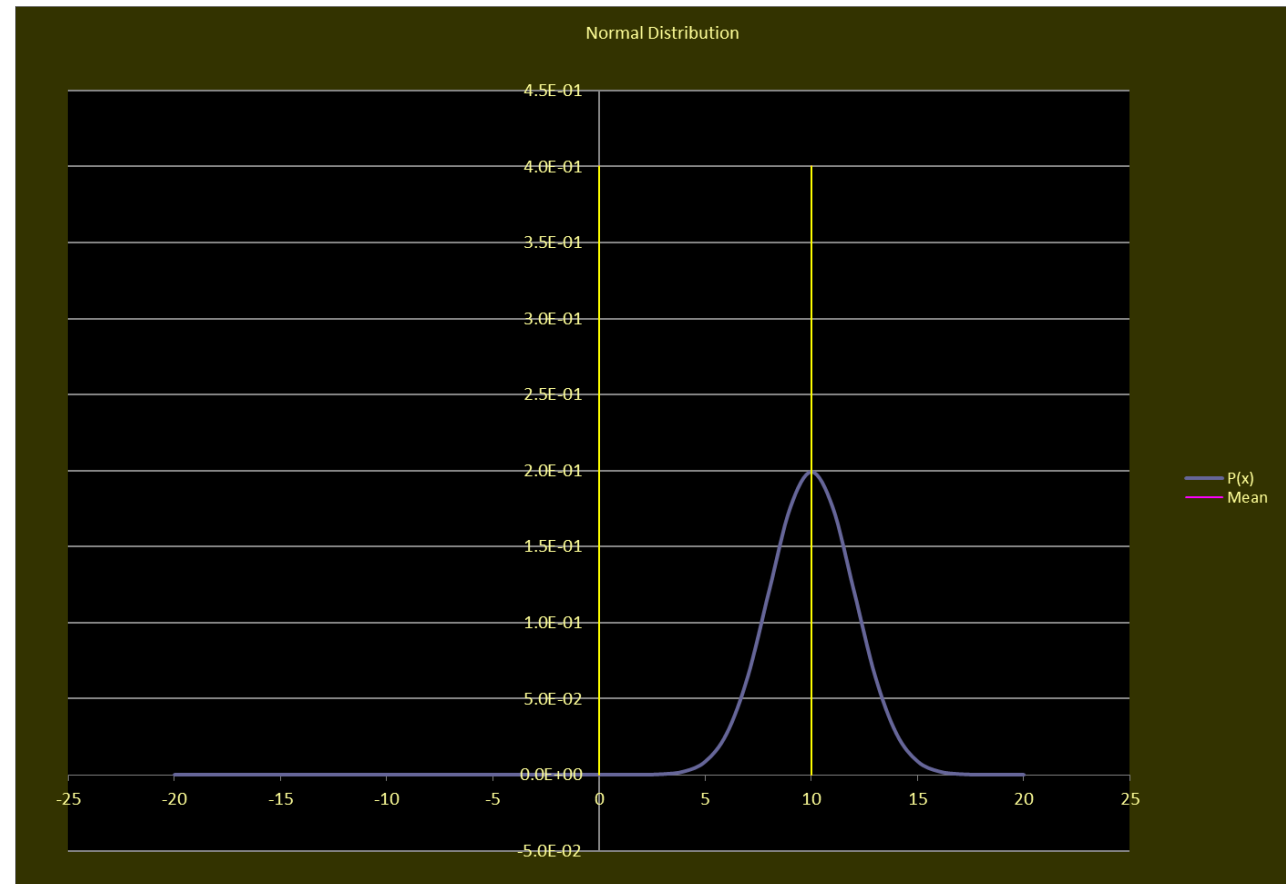
Normal Distribution

Mean = 5; Standard Deviation = 2



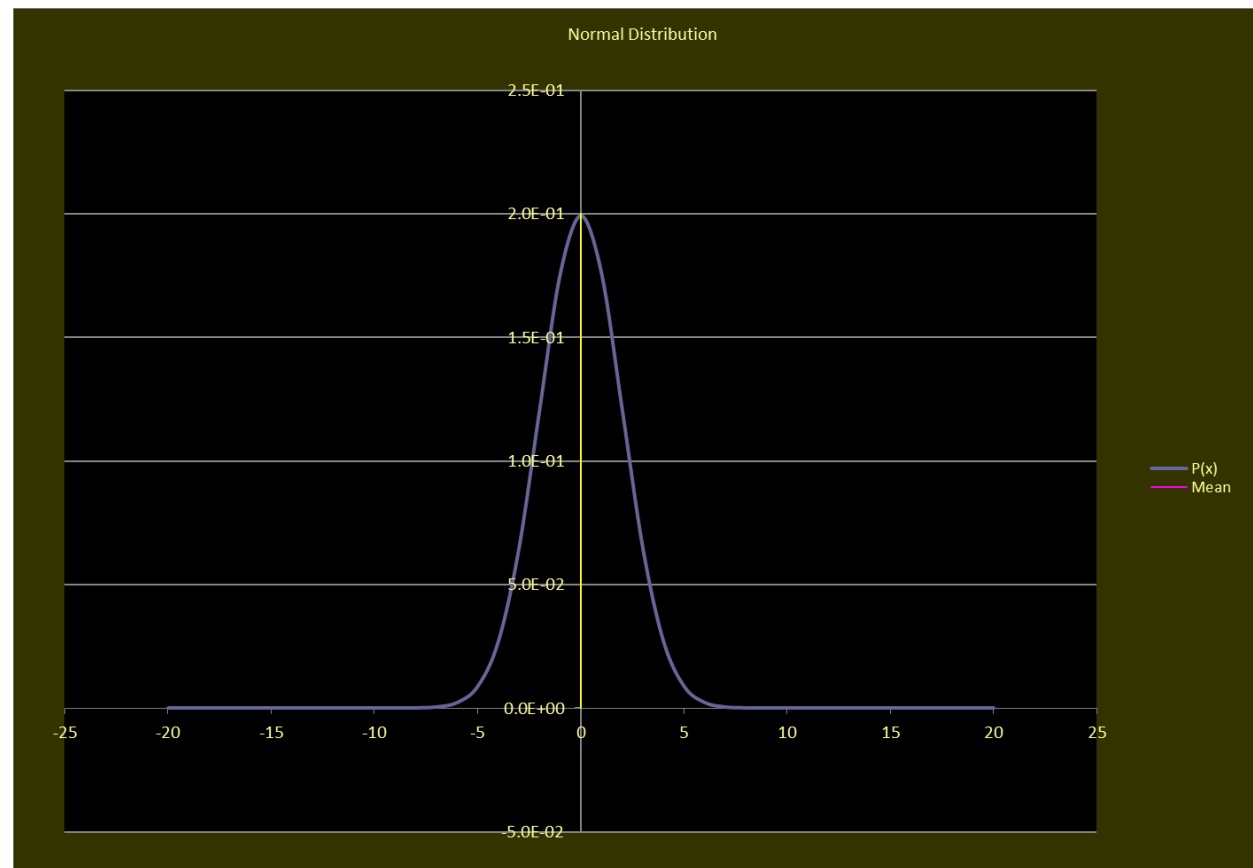
Increase mean of Normal Distribution

Mean = **10**; Standard Deviation = 2



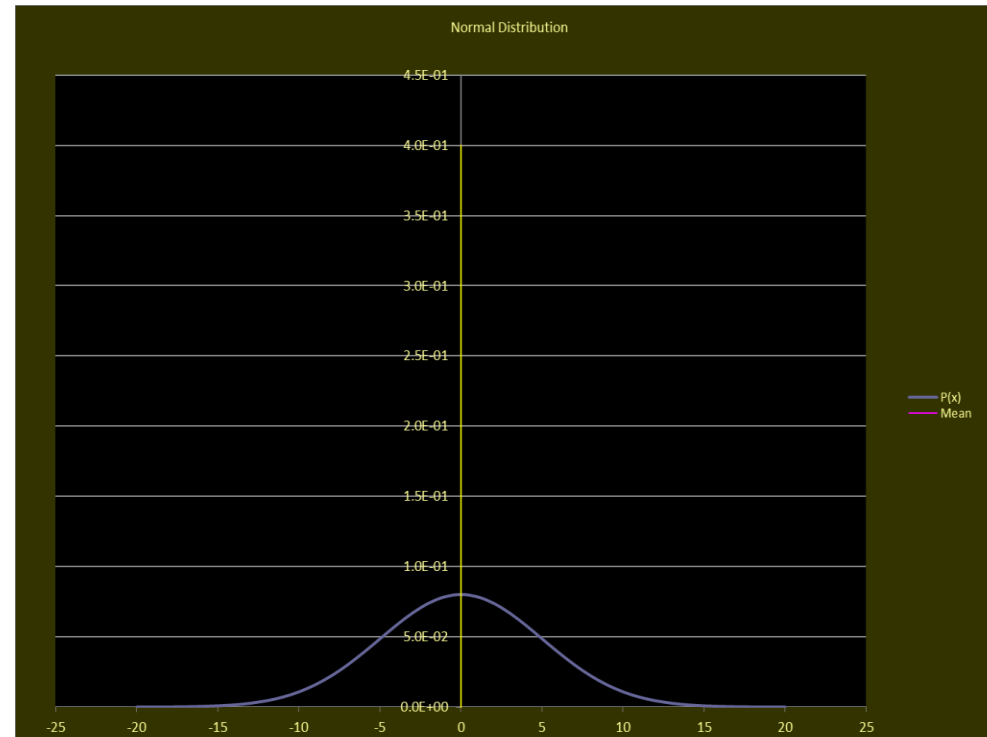
Decrease mean of Normal Distribution

Mean = 0; Standard Deviation = 2



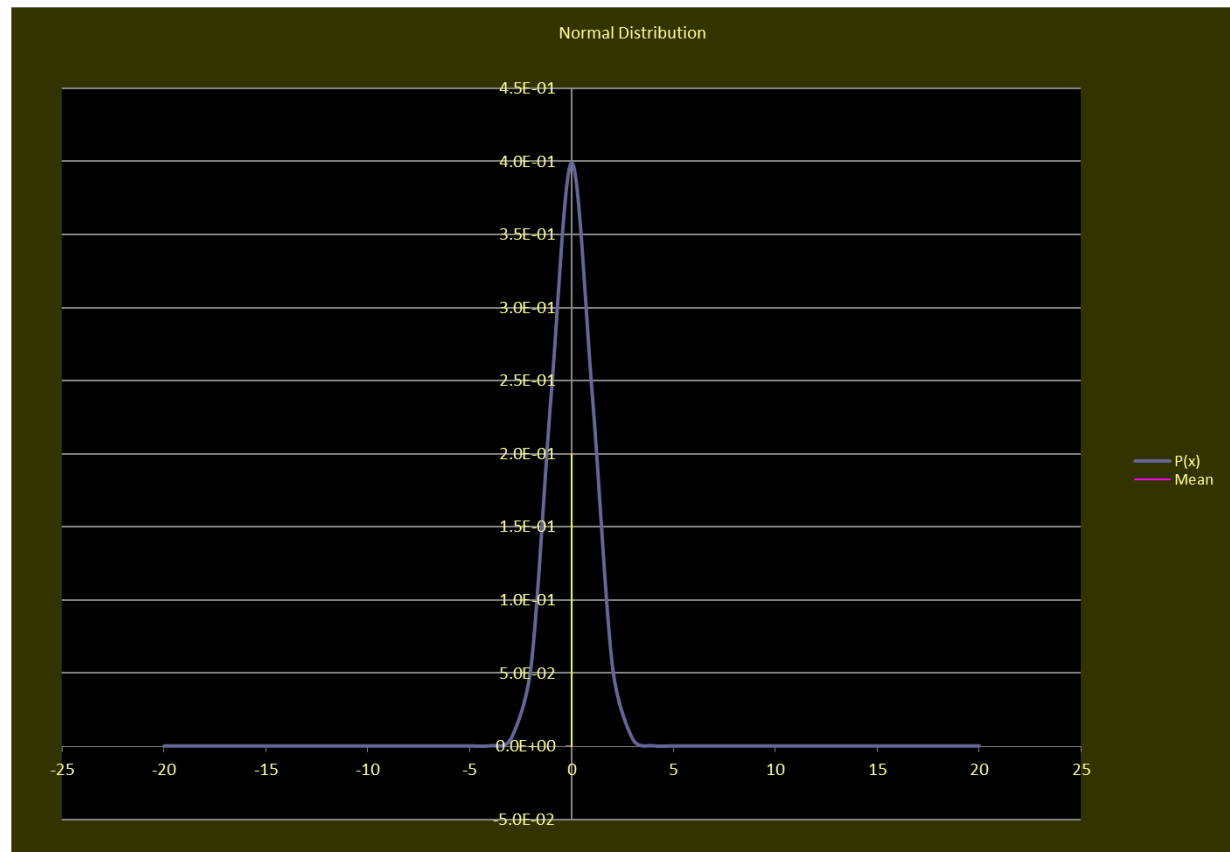
Increase SD of Normal Distribution

Mean = 0; Standard Deviation = 5

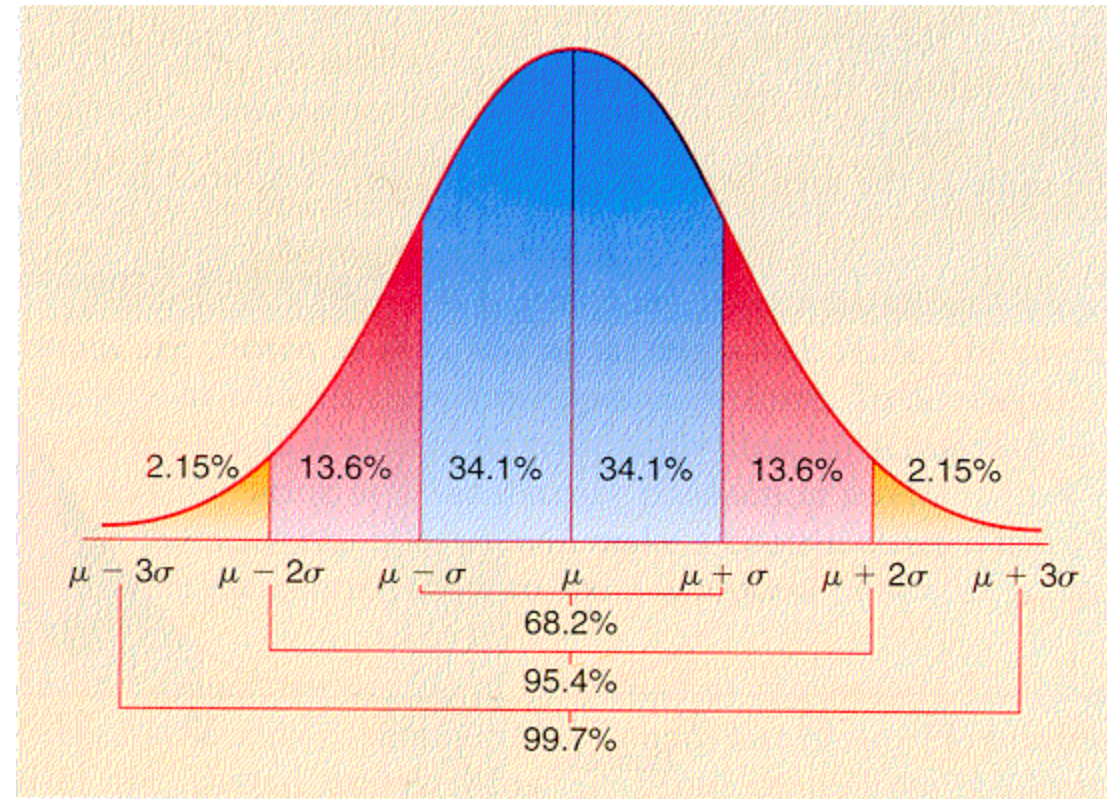


Decrease SD of Normal Distribution

Mean = 0; Standard Deviation = 1

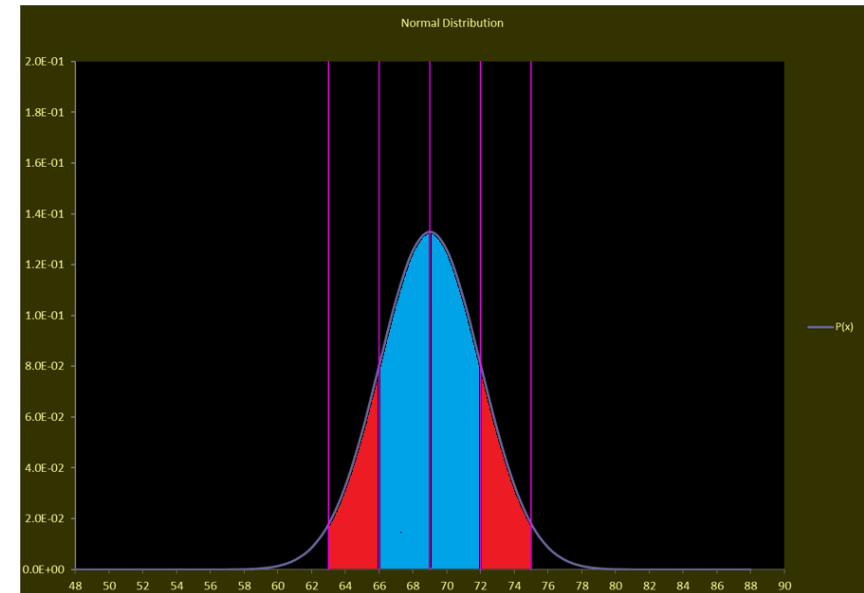


Some applications of normal distribution



Distribution of heights of Indian Men

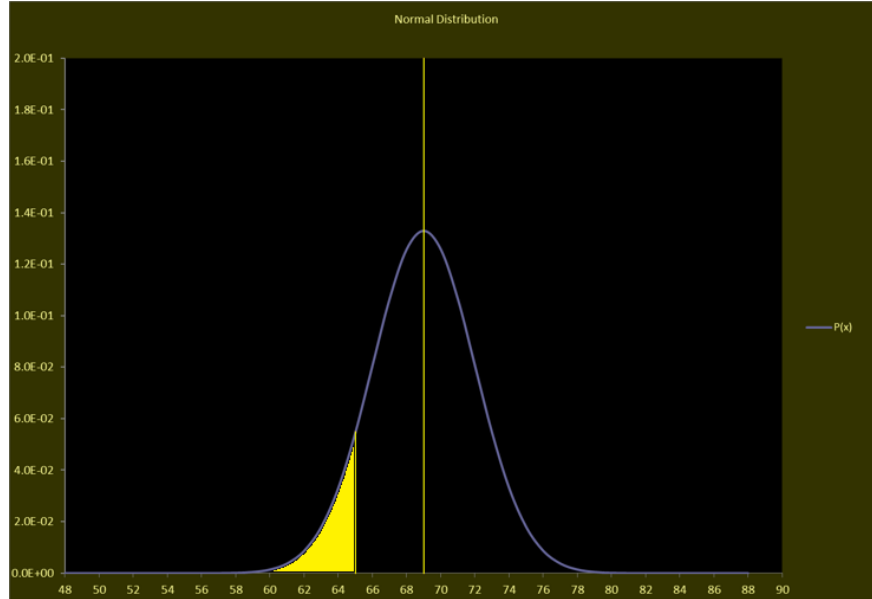
Average = 69 inches; SD = 3 inches



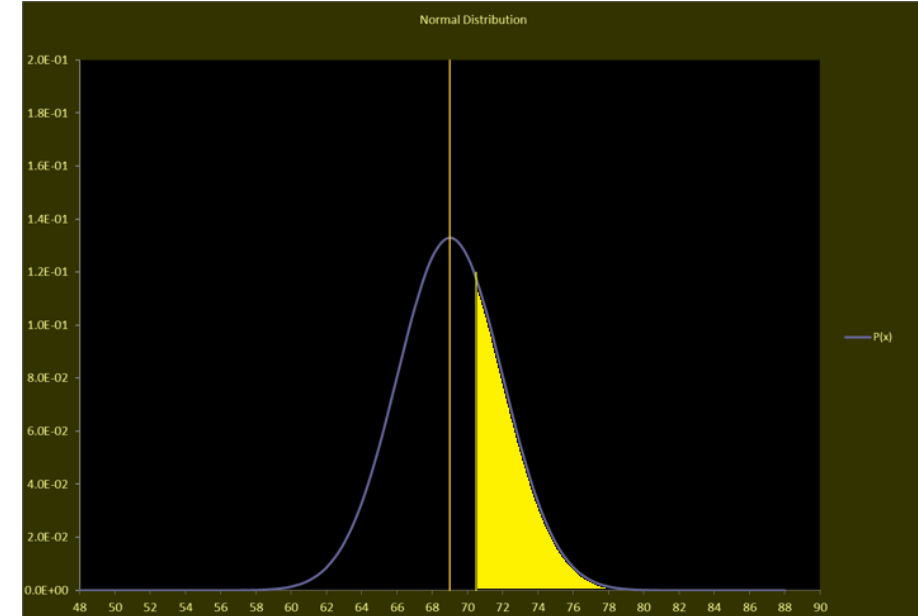
Normal distribution implies: 68.2% of Indian men are between 66 inches and 72 inches
95.4% of Indian men are between 63 inches and 75 inches

More information

What proportion of Indian men are shorter than 65 inches?



What proportion of Indian men are taller than 70.5 inches?



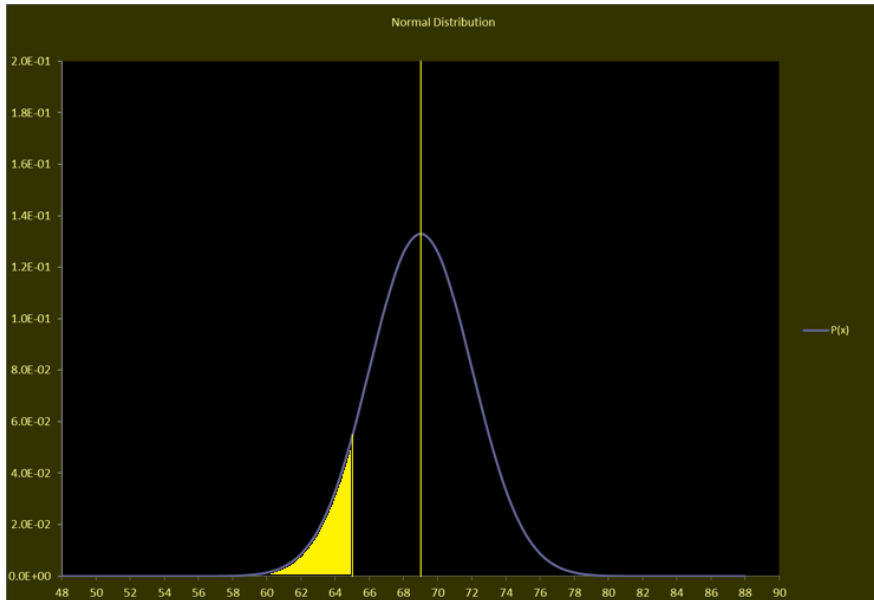
Functions for the normal distribution

$\text{NORM.DIST}(x, \mu, \sigma, \text{true}) = P(X \leq x)$. This gives the **left-tail** probability for mean μ and standard deviation σ .

$\text{NORM.INV}(\text{prob}, \mu, \sigma) = x$ such that $P(X \leq x) = \text{prob}$. This gives the value of x that places the specified probability in the **left tail**.

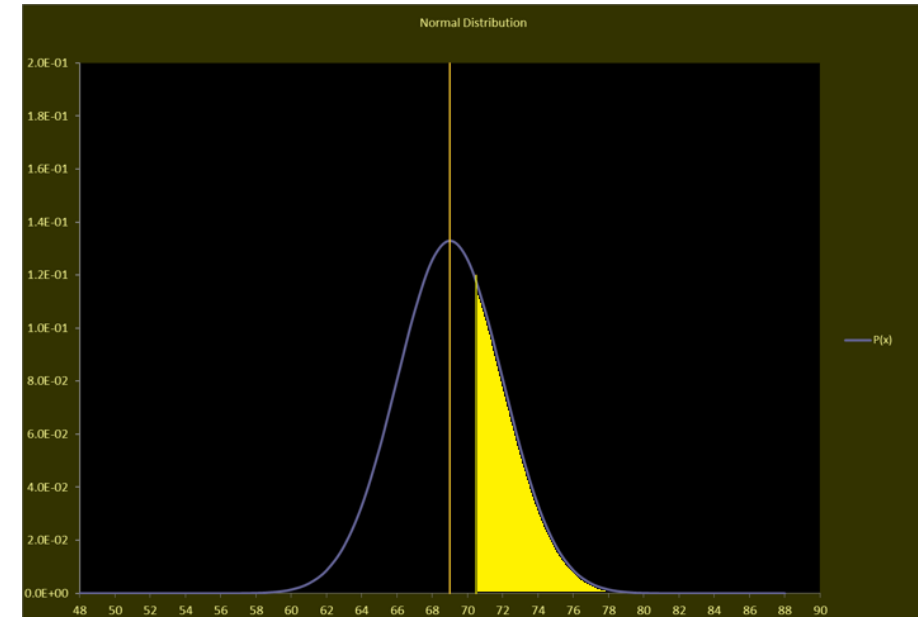
More information

What proportion of Indian men are shorter than 65 inches?



9.12% of Indian men are shorter than 65 inches

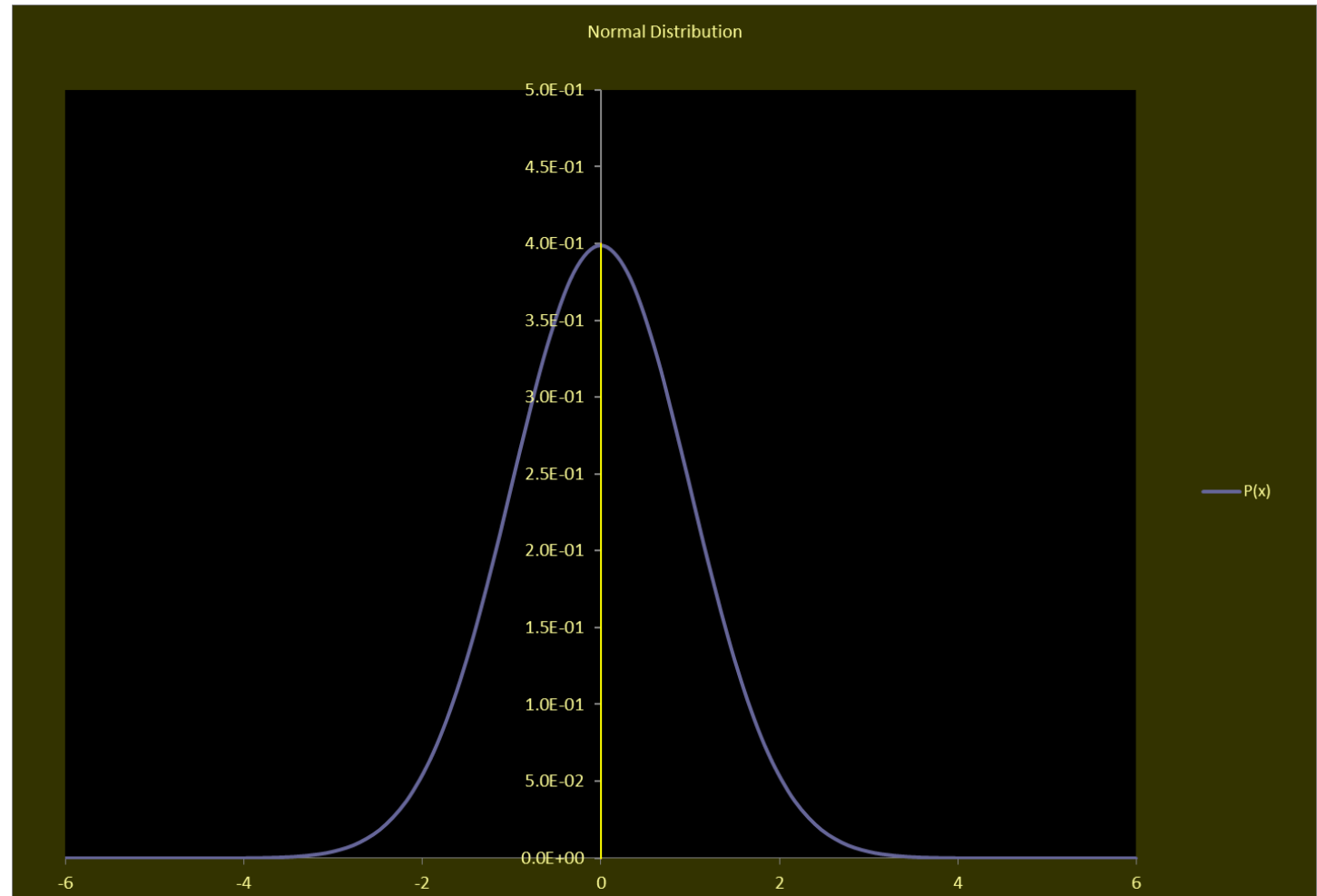
What proportion of Indian men are taller than 70.5 inches?



30.85% of Indian men are taller than 70.5 inches

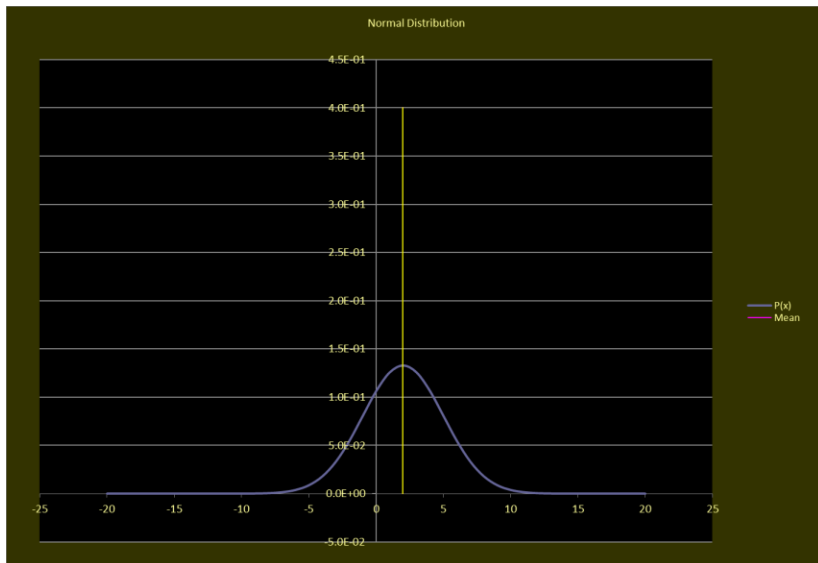
Standard Normal Distribution

- Mean = 0; SD = 1

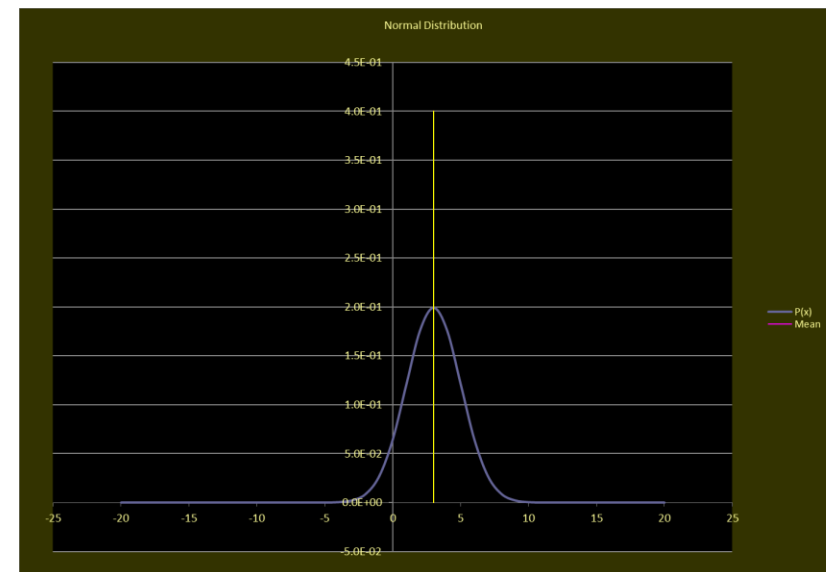


Why do we care about Standard Normal Distribution?

Example: Consider stock returns of two companies



Stock 1:
Mean = 30%
SD = 20%



Stock 2:
Mean = 20%
SD = 30%

How do we compare these two stocks? For example, what is the likelihood of receiving *at least* 40% return in each of the stocks?

Z-score

Definition: Z-score is the number of standard deviations away from the mean that a certain data point is.

Calculation of Z-score: $Z = (x - \text{Mean}) / \text{SD}$

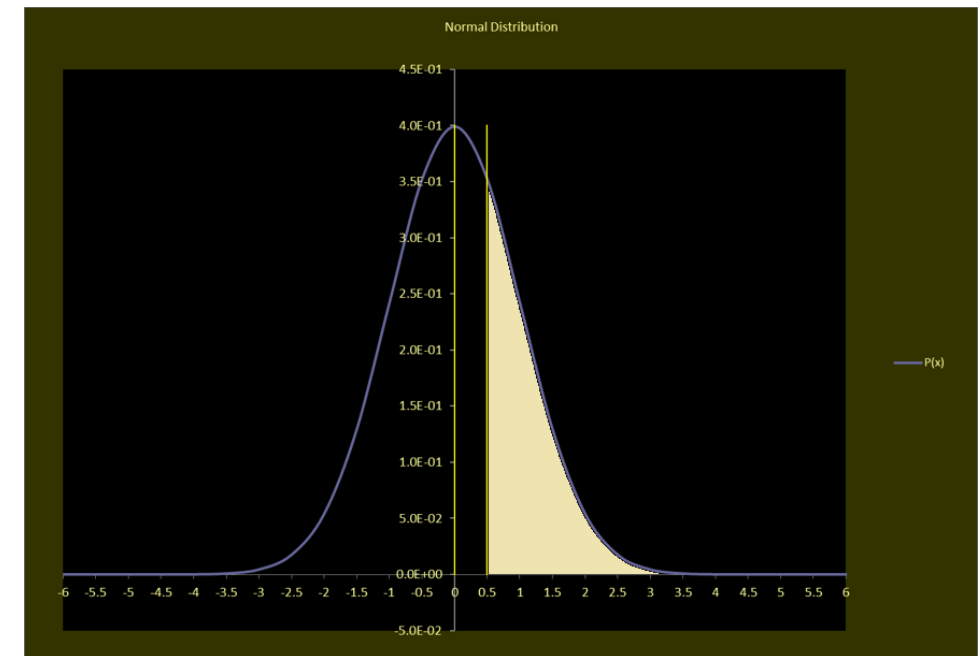
What is the Z-score for stock 1 corresponding to 40% return?

Answer: $Z = (x - \text{Mean}) / \text{SD} = (40 - 30) / 20 = 0.5$

Likelihood of getting at least 40% return =
Probability ($Z > 0.5$)

Using standard tables (or Excel functions), we can find that likelihood of

receiving at least 40% return with stock 1 is
30.1%



What is the Z-score for stock 2 corresponding to 40% return?

Ans: $Z = (x - \text{Mean})/\text{SD} = (40-20)/30 = 0.66$

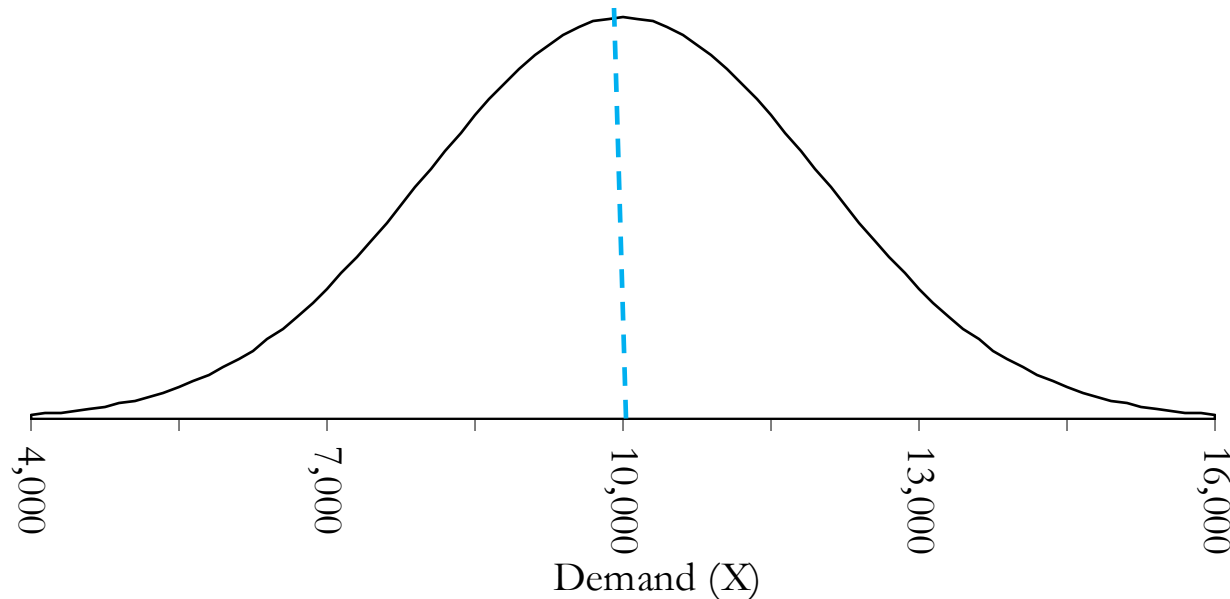
Using standard tables (or Excel functions), we can find that likelihood of receiving at least 40% return with stock 2 is 25.5%

NORMAL DISTRIBUTION: EXAMPLE

Example – Halloween Costume

- Demand for a Halloween costume is expected to be normally distributed with a mean demand of 10,000 and a standard deviation of 2000 units.

Demand Distribution (Mean $\mu=10,000$, Std $\sigma=2000$)



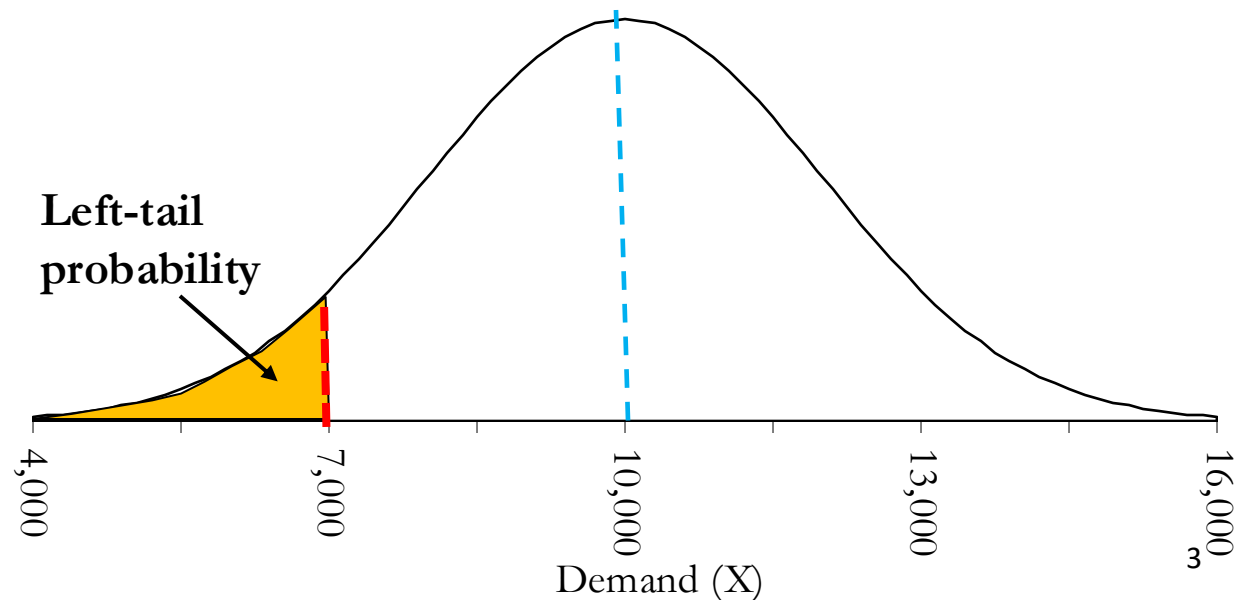
Questions

a) What is the probability that the demand will be less than or equal to 7,000 units?

➤ $\Pr(X \leq 7,000) = ?$

Left-tail probability

Demand Distribution (Mean $\mu=10,000$, Std $\sigma=2000$)



a) What is the probability that the demand will be less than or equal to 7,000 units?

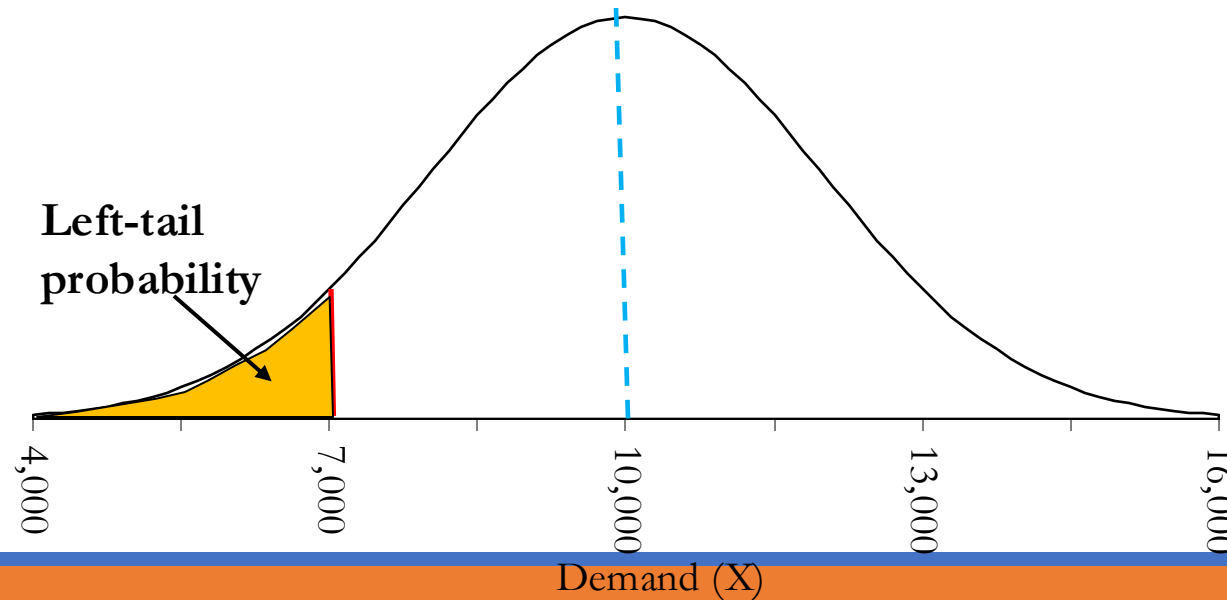
$$\Pr (X \leq 7,000) = \text{NORM.DIST} (7000, 10000, 2000, \text{TRUE})$$
$$= 0.0668$$

Left-tail
probability

Mean μ

Std σ

Demand Distribution (Mean $\mu=10,000$, Std $\sigma=2000$)

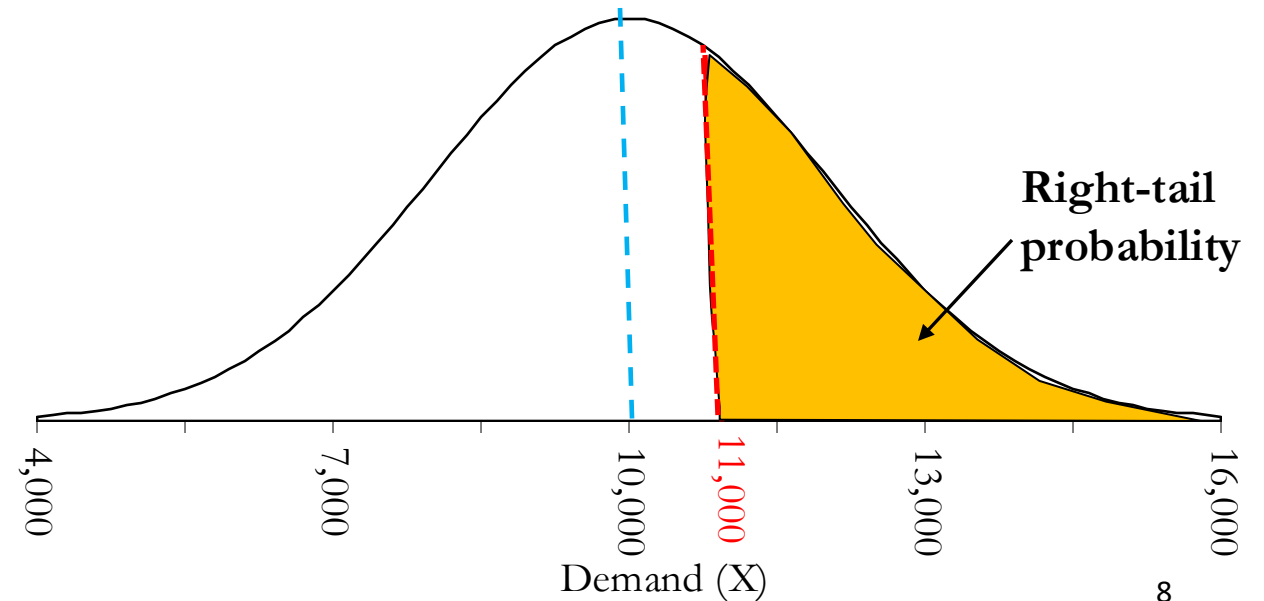


Questions

b) If the retailer stocks 11,000 units of this item, what is the chance of selling out all units?

- All units will be sold when Demand > 11,000
- $\Pr(X > 11,000) = ?$

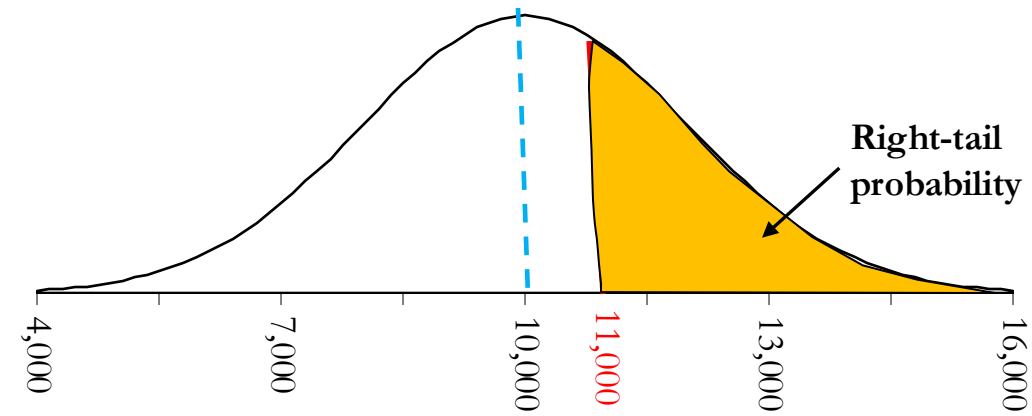
Demand Distribution (Mean $\mu=10,000$, Std $\sigma=2000$)



c) What is the probability that the demand exceeds 11,000 units?

$$\begin{aligned}\Pr (X > 11,000) &= 1 - \text{NORM.DIST} (11000, 10000, 2000, \text{TRUE}) \\ &= 0.3085\end{aligned}$$

Demand Distribution (Mean $\mu=10,000$, Std $\sigma=2000$)

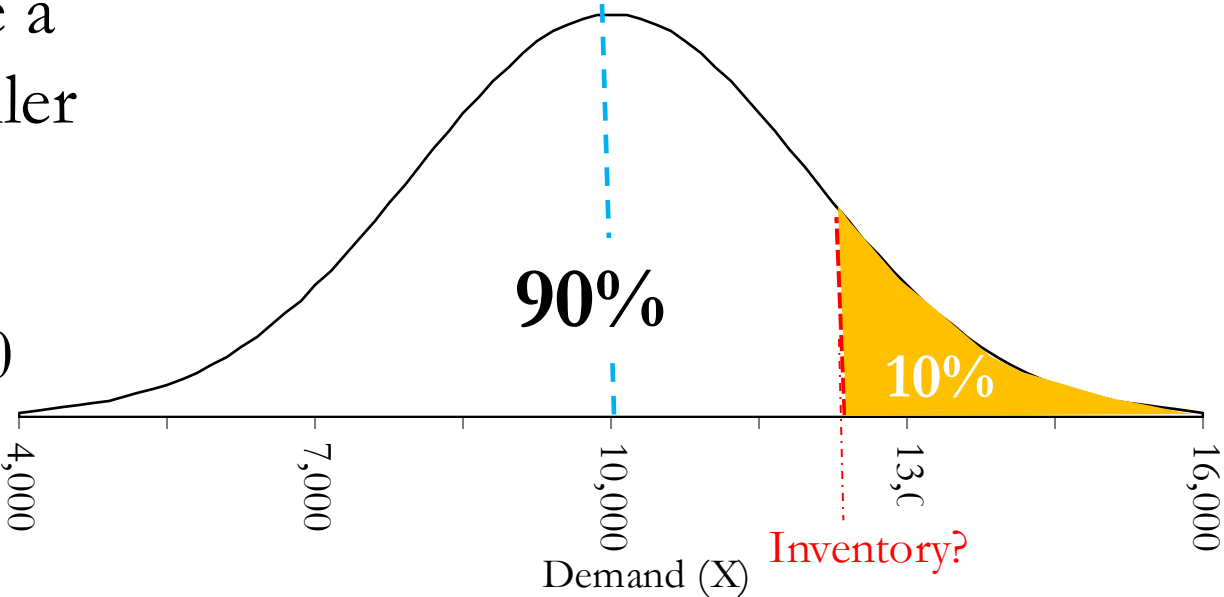


Questions

c) The retailer wants to keep the stock-out risk at 10%. What inventory level would achieve this goal?

➤ When demand exceeds the inventory level, the retailer will experience a stock-out. That means, the retailer wants

Demand Distribution (Mean $\mu=10,000$, Std $\sigma=2000$)



To be determined

$$\Pr (\text{Demand} > \text{Inventory}) = 0.10$$

Inventory =

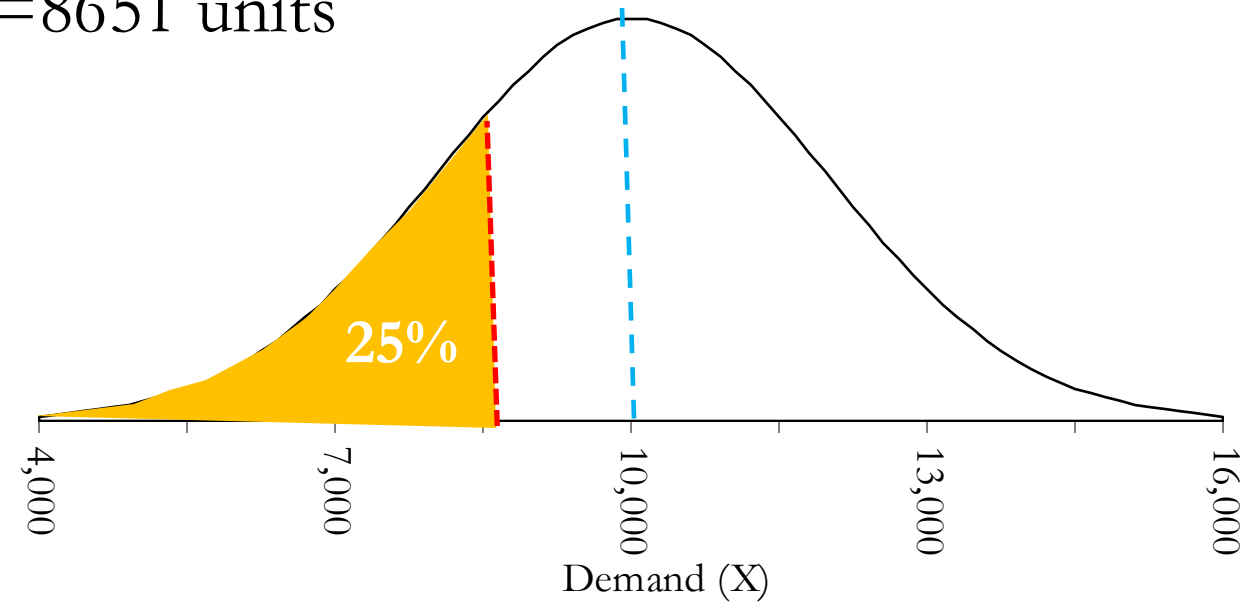
$$\text{NORM.INV}(0.90, 10000, 2000) = 12563 \text{ units}$$

Questions

d) Demand is smaller than X with a 25% probability. What X value will make this statement correct?

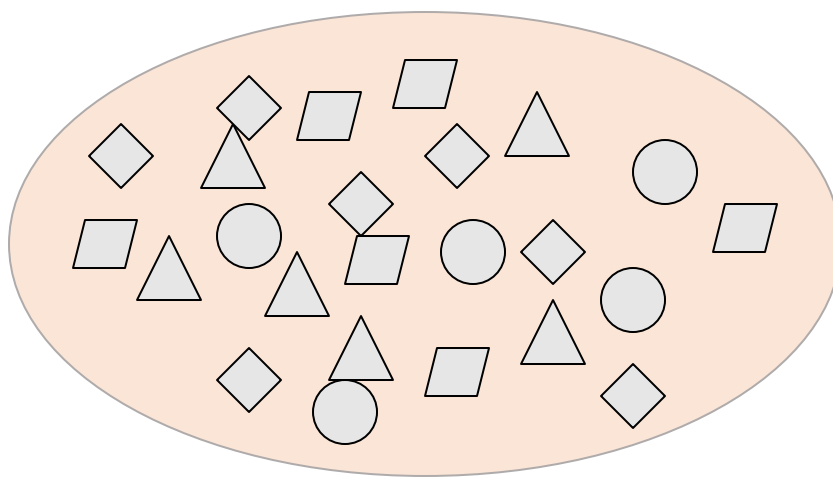
Demand Distribution (Mean $\mu=10,000$, Std $\sigma=2000$)

$$\text{NORM.INV}(0.25, 10000, 2000) = 8651 \text{ units}$$



Populations, Samples, and Summary Measures

Populations vs. Samples



Population

Sample

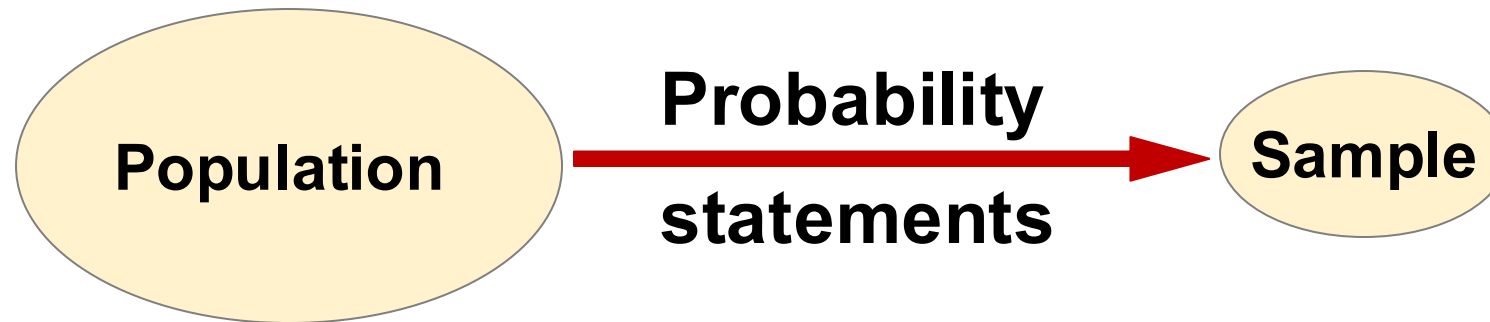
Populations vs. Samples

- **Population:** the total collection of items that fall within the scope of the statistical investigation
- **Sample:** some subset of the population

Probability Model

Characteristics of the population are known.

We want to make statements about the various types of samples which we might obtain.

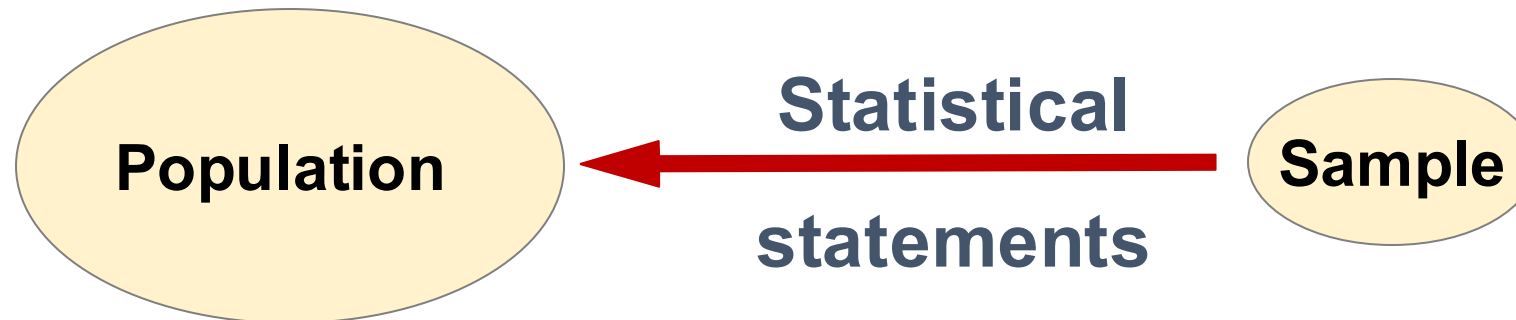


- “From the general to the specific”
- Not always possible to know the population (but we may *assume* we know)

Statistical Model

We only have the sample available.

We want to make statements (inferences) about the population.



- “From the specific to the general”
- It’s always possible to take a sample

Summary measures: calculated numbers that characterize a set of values:

- Population parameters
- Sample statistics

P_{opulation} P_{arameters}

- Are constant, but unknown
- Mean μ
- Standard deviation σ
- Proportion (probability of success) π



Let there be **N** elements in the population.
Mother Nature can calculate *population parameters*.
All sums are taken over all N observations.

Population mean: $\mu = \frac{\sum x_i}{N} = \frac{x_1 + x_2 + \dots + x_N}{N}$

Population variance: $\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$
 $= \frac{(x_1 - \mu)^2 + (x_2 - \mu)^2 + \dots + (x_N - \mu)^2}{N}$

Population standard deviation: $\sigma = \sqrt{\sigma^2}$

Sample Statistics

- Are known but variable (they depend on the sample)
- Mean $\bar{X}$
- Standard deviations (s)



Let there be **n** elements in the sample.

We can calculate *sample statistics*.

All sums are taken over all n observations.

$$\text{Sample mean: } \bar{X} = \frac{\sum x_i}{n} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

$$\begin{aligned} \text{Sample variance: } s^2 &= \frac{\sum (x_i - \bar{X})^2}{n-1} \\ &= \frac{(x_1 - \bar{X})^2 + (x_2 - \bar{X})^2 + \dots + (x_n - \bar{X})^2}{n-1} \end{aligned}$$

$$\text{Sample standard deviation: } s = \sqrt{s^2}$$

What's the point of a sample?

Why do we take samples?

So we can use the sample statistics to estimate the corresponding population parameters

sampling

μ

$\bar{x}$



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THANK YOU!